EXP-8 IMAGE GENERATION USING VARIATIONAL AUTOENCODER (VAE)

AIM:

To Implement a variational Autoencoder that learns latent representation of images and generates new images from the learned distribution

SOFTWARE REQUIREMENT:

* Python 3.x
* Pytorch
* Torchvision
* Matplotlib
* CelebA Dataset

ALGORITHM:

* Import the required libraries
* Load and preprocess the image dataset
* Construct an encoder to extract latent representations
* Calculate the mean and log variance
* Apply the reparameterization technique
* Construct a decoder to reconstruct images
* Calculate reconstruction loss and KL divergence
* Train the VAE using the Adam optimizer
* Generate images from random latent vectors
* Display the generated images.

PROGRAM:

```
import torch
import torch.nn as nn
import torch.nn.functional as F
import matplotlib.pyplot as plt.
```

```
class VAE(nn.Module):
    def __init__(self, latent_dim=100):
        super().__init__()
        self.encoder = nn.Sequential(
            nn.Flatten(),
            nn.Linear(3*64*64, 512),
            nn.ReLU()
        )
        self.mu = nn.Linear(512, latent_dim)
        self.logvar = nn.Linear(512, latent_dim)

    ) ~~self.decoder = nn.Sequential(~~

    def forward(self, x):
        h = self.encoder(x)
        mu = self.mu(h)
        logvar = self.logvar(h)

        std = torch.exp(0.5*logvar)
        z = mu + torch.randn_like(std)*std


        output = self.decoder(z)
        output = output.view(-1, 3, 64, 64)

        return output, mu, logvar

def vae_loss(recon, x, mu, logvar):
    reconstruction = F.mse_loss(
        recon, x, reduction="sum"
    )
```

EXPECTED OUTPUT

VAE model created successfully.

```
VAE (
    (encoder): Sequential(...)
    (mu): Linear(...)
    (logvar): Linear(...)
    (decoder): Sequential(...)
)
```

```
    KL = -0.5 * torch.sum(
        1 + logvar - mu.pow(2) - logvar.exp()
    )
    return reconstruction + KL

device = torch.device(
    "cuda" if torch.cuda.is_available() else "cpu")

model = VAE().to(device)
optimizer = torch.optim.Adam(
    model.parameters(), lr=0.001
)
print("VAE model created successfully")
print(model)
```

RESULT:

The variational Autoencoder architecture was implemented. It learns a latent representation through an encoder and reconstructs image through a decoder. New images can be generated by sampling from the latent space after training.